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A Survey on Large Language Model-based Multi-Agent Systems: Paradigms, Applications, and Challenges

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Abstract

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Keywords

computing and processing

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Abstract

With the rapid advancement of Large Language Models (LLM), artificial intelligence is undergoing a paradigm shift from solitary intelligence to collaborative Multi-Agent Systems (MAS). By simulating the collaborative division of labor found in human societies, LLM-based Multi-Agent Systems (LLM-MAS) effectively overcome the performance bottlenecks inherent in single-agent models when addressing complex tasks. Addressing the prevailing fragmentation and the absence of a unified taxonomy in current research, this paper introduces a three-dimensional analytical framework integrating Structure, Interaction, and Evolution. We systematically elucidate the evolutionary logic of organizational architectures from static presets to dynamic adaptation; deeply dissect the underlying mechanisms of communication paradigms and collective decision-making; and explore pathways to self-evolution through prompt restructuring, memory consolidation, and parameter updates. Furthermore, we categorize cutting-edge applications into three domains: “AI for Humans”, “AI for the World”, and “AI for AI”. Finally, we analyze critical challenges, including hallucination propagation and safety governance, providing clear theoretical support and a technical roadmap for advancing the field toward artificial general intelligence.

1 Introduction

With the advancement of Large Language Models (LLM) in understanding and reasoning capabilities, artificial intelligence is transitioning from solitary intelligence to collaborative Multi-Agent Systems (MAS). Although single-agent models are broadly capable, they are still limited by finite context windows. They also suffer from blind spots that come from relying on a single perspective, especially for long-horizon planning, complex reasoning, and cross-domain knowledge integration. In contrast, LLM-based multi-agent

systems (LLM-MAS) break a task into manageable subproblems and assign them to specialized roles. Through complementary expertise and cross-checking, the system can outperform a single agent [Wang *et al.*, 2025a]. Currently, LLM-MAS have demonstrated significant potential and application value in complex tasks such as software development [Seo *et al.*, 2025], social simulation [Piao *et al.*, 2025], scientific research [Weng *et al.*, 2025a], and GUI interaction [Zhou *et al.*, 2025]. These advances also make LLM-MAS a critical pathway toward higher-level Artificial General Intelligence.

Current research on multi-agent systems centers on three core dimensions: organizational structure, interaction mechanisms, and evolutionary strategies. Early explorations primarily relied on heuristic-based static Standard Operating Procedures (SOPs) to ensure stability through deterministic linear flows. Complex and long-horizon tasks demand greater adaptability, and recent work reflects this shift. System designs are moving from fixed orchestration to dynamic topologies [Li *et al.*, 2025b], from explicit natural-language exchanges to more efficient implicit vector communication [Fu *et al.*, 2025], and from prompt-only tuning to learning and memory-based updates [Chen *et al.*, 2025b]. These advancements aim to overcome the rigidity of traditional architectures, thereby enhancing the system’s adaptability and generalization capabilities in complex environments.

The explosive growth of LLM-MAS literature has yielded a rich array of solutions in specific application domains. Navigating this complex landscape requires distilling the core design-level paradigms that transcend specific tools and frameworks. By synthesizing these diverse developments into a unified perspective, we can better understand the system lifecycle through the critical “Structure-Interaction-Evolution” technical thread, establishing a solid theoretical foundation for the field. To this end, this paper aims to provide a comprehensive survey of LLM-MAS. As illustrated in Figure 1, we propose a three-dimensional analytical framework integrating “Structure-Interaction-Evolution”. The framework traces how architectures evolve from static designs to dynamic ones. It also examines communication paradigms and decision-making mechanisms, and summarizes how systems iterate through prompts, memory, and parameters. Beyond technical implementation, this paper fo-

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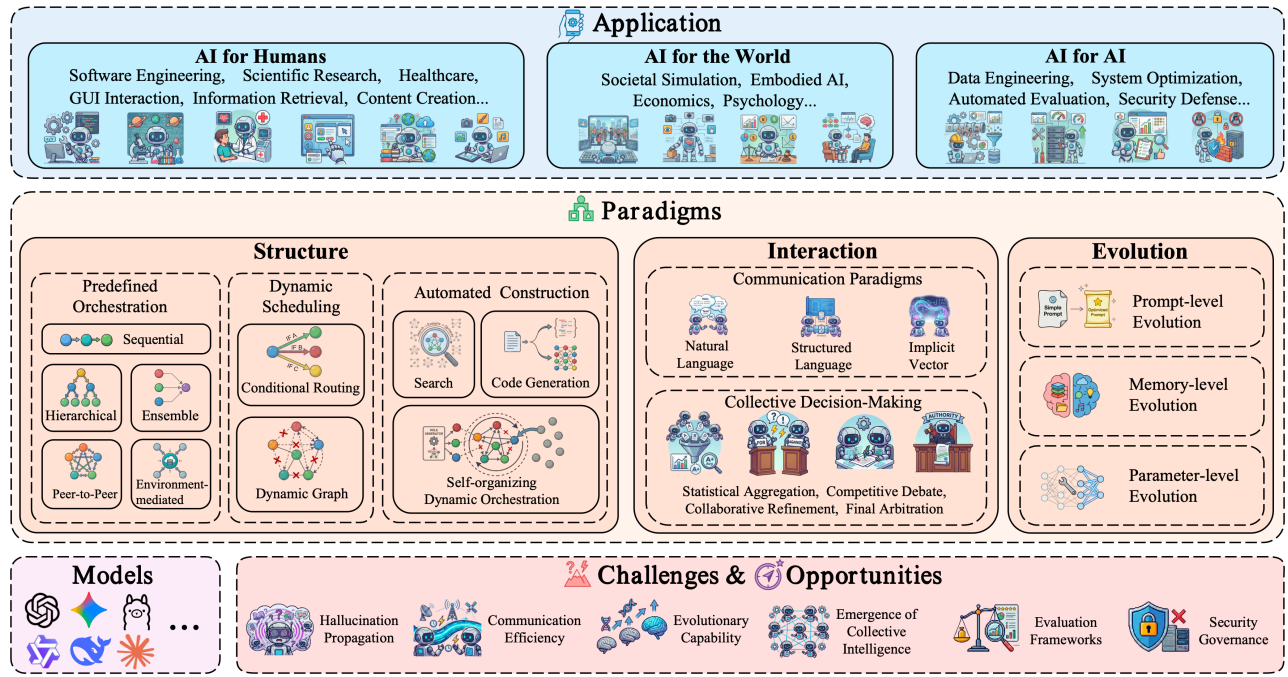


Figure 1: Overview of the proposed three-dimensional analytical framework for LLM-MAS. The figure illustrates how this survey organizes existing work along three core dimensions—Structure, Interaction, and Evolution. The middle panel summarizes representative paradigms within each dimension, including organizational structures, communication and decision-making mechanisms, and evolutionary strategies. The top panel categorizes frontier applications into AI for Humans, AI for the World, and AI for AI, while the bottom panel highlights key challenges and opportunities that span the entire lifecycle of LLM-MAS.

cuses on analyzing the trade-offs and applicable scenarios behind different design choices, aiming to provide researchers with clear technical navigation and theoretical support.

The remainder of this paper is organized as follows: Section 2 details the core paradigms of multi-agent collaboration, systematically classifying them across the dimensions of structure, interaction, and evolution; Section 3 categorizes frontier applications into “AI for Humans”, “AI for the World”, and “AI for AI”; Section 4 delves into critical challenges and opportunities, including hallucination propagation, communication efficiency, evolutionary capabilities, and safety governance; and Section 5 concludes the paper.

2 Multi-Agent Collaboration Paradigms: Structure, Interaction, and Evolution

2.1 Structure

This section focuses on how agent roles are assigned and how collaboration structures are designed in LLM-MAS.

Predefined Orchestration

Predefined orchestration involves manually setting role division and interaction topology before deployment. The structure remains unchanged during execution, trading process certainty for stability and reproducibility. As a result, this paradigm is suitable for standardized scenarios with clear task flows and minimal variation.

Sequential. This approach uses a chain-like pipeline structure to decompose complex tasks, reducing cognitive load.

[Hong *et al.*, 2024] established a waterfall paradigm based on Standard Operating Procedures (SOPs), and recent research has expanded this paradigm to fields such as scientific research [Seo *et al.*, 2025]. While this architecture sacrifices some flexibility, it ensures execution stability and reproducibility for standardized tasks.

Hierarchical. This employs a tree topology to build a “Manager-Worker” vertical hierarchy, where top-level nodes are responsible for planning and coordinating the parallel work of lower-level specialized agents. Related research, such as [Agashe *et al.*, 2025] and [Hegazy *et al.*, 2025], effectively mitigates long-horizon planning and decision convergence difficulties in complex tasks through fine-grained vertical division of labor and hierarchical arbitration.

Ensemble. Ensemble structures utilize parallel reasoning by multiple agents to acquire cognitive diversity, and then integrate multi-perspective responses to eliminate individual hallucinations. [Wang *et al.*, 2025a] and [Tang *et al.*, 2025] demonstrate that through the integration of heterogeneous open-source models, systems can exhibit powerful emergent collective intelligence, effectively improving robustness and accuracy in complex tasks like scientific reasoning.

Peer-to-Peer. Peer-to-peer structures remove scheduling-type central nodes, forming a decentralized collaborative mode. Debate-style systems like [Liang *et al.*, 2024] and [Li *et al.*, 2025a] can be viewed as result aggregation following peer execution. This paradigm is limited by communication

Work	Modality	Structure	Communication Paradigm	Collective Decision-Making				Evolution	Open Source
				SA	CR	CD	FA		
[Tian <i>et al.</i> , 2025]	Text	Automated (Self-organizing)	Natural language	-	✓	-	✓	Prompt	Yes
[Agashe <i>et al.</i> , 2025]	Text, Vision	Predefined (Hierarchical)	Structured language	-	✓	-	-	-	Yes
[Wang <i>et al.</i> , 2025d]	Text	Dynamic (Dynamic graph)	Natural language	-	✓	-	-	-	Yes
[Ma <i>et al.</i> , 2025]	Text	Automated (Self-organizing)	Natural language	✓	✓	-	-	Prompt	-
[Liao <i>et al.</i> , 2025]	Text, Vision, Audio	Predefined (Hierarchical)	Structured language	-	✓	-	-	-	-
[Zhang <i>et al.</i> , 2025d]	Text	Dynamic (Dynamic graph)	Natural language	-	✓	✓	-	-	Yes
[Piao <i>et al.</i> , 2025]	Text	Predefined (Environment-mediated)	Structured language	-	-	-	-	Memory	Yes
[Li <i>et al.</i> , 2025b]	Text	Automated (Self-organizing)	Natural language	✓	✓	-	-	-	Yes
[Rong <i>et al.</i> , 2025]	Text, Vision, Audio	Dynamic (Conditional Routing)	Structured language	-	✓	-	-	-	Yes
[Zhang <i>et al.</i> , 2025f]	Text	Predefined (Hierarchical)	Natural language	-	✓	✓	✓	-	-
[Fu <i>et al.</i> , 2025]	Text	Predefined (Any)	Implicit vector	-	-	-	-	-	Yes
[Choi <i>et al.</i> , 2025]	Text	Predefined (Peer-to-peer)	Natural language	✓	-	✓	-	-	Yes
[Hu <i>et al.</i> , 2025b]	Text	Automated (Self-organizing)	Natural language	-	✓	-	-	Prompt	Yes
[Fan <i>et al.</i> , 2025]	Text	Predefined (Sequential)	Natural language	-	✓	-	✓	Memory	-
[Zhang <i>et al.</i> , 2025b]	Text	Predefined (Environment-mediated)	Natural language	-	✓	-	-	Memory	Yes
[Takano <i>et al.</i> , 2025]	Text	Predefined (Peer-to-peer)	Natural language	-	-	✓	-	-	-
[Bi <i>et al.</i> , 2026]	Text	Predefined (Ensemble)	Natural language	✓	✓	✓	-	-	-
[Ye <i>et al.</i> , 2025a]	Text	Predefined (Any)	Implicit vector	-	-	-	-	-	Yes
[Heng <i>et al.</i> , 2025]	Text	Predefined (Hierarchical)	Structured language	-	✓	-	-	Parameter	Yes
[Feng <i>et al.</i> , 2025]	Text	Predefined (Peer-to-peer)	Natural language	-	-	✓	-	-	Yes
[Zhang <i>et al.</i> , 2025c]	Text	Automated (Search)	Natural language	✓	✓	✓	-	Prompt	Yes
[Qian <i>et al.</i> , 2025]	Text	Predefined (Peer-to-peer)	Natural language	-	✓	-	-	-	Yes
[Park <i>et al.</i> , 2025]	Text	Predefined (Peer-to-peer)	Natural language	✓	✓	-	-	Parameter	Yes
[Ye <i>et al.</i> , 2025b]	Text	Automated (Code-generation)	Natural language	-	✓	-	✓	-	Yes
[Wang <i>et al.</i> , 2025b]	Text	Automated (Self-organizing)	Structured language	-	✓	-	-	Prompt	Yes
[Zhang <i>et al.</i> , 2025h]	Text	Automated (Code-generation)	Natural language	-	✓	-	-	-	Yes
[Wang <i>et al.</i> , 2025a]	Text	Predefined (Ensemble)	Natural language	-	✓	-	-	-	Yes
[Wang <i>et al.</i> , 2025c]	Text, Vision	Predefined (Hierarchical)	Structured language	-	✓	-	-	Memory	Yes
[Xiao <i>et al.</i> , 2025]	Text	Predefined (Peer-to-peer)	Natural language	-	-	✓	✓	-	-
[Chen <i>et al.</i> , 2025b]	Text	Predefined (Sequential)	Natural language	-	✓	-	✓	Parameter	Yes
[Seo <i>et al.</i> , 2025]	Text	Predefined (Sequential)	Structured language	-	✓	-	-	-	Yes
[Agrawal <i>et al.</i> , 2025]	Text	Automated (Code-generation)	Natural language	-	✓	-	✓	-	Yes
[Zhao <i>et al.</i> , 2025]	Text	Predefined (Sequential)	Natural language	-	✓	-	-	Parameter	-
[Li <i>et al.</i> , 2025a]	Text	Predefined (Peer-to-peer)	Structured language	-	-	✓	✓	-	Yes
[Yamada <i>et al.</i> , 2025]	Text, Vision	Predefined (Hierarchical)	Structured language	-	✓	-	✓	Memory	Yes
[Swanson <i>et al.</i> , 2025]	Text	Predefined (Hierarchical)	Structured language	-	✓	-	✓	Memory	Yes

Table 1: Comparison of representative works related to LLM-MAS across structure, interaction, and evolution. SA/CR/CD/FA denote Statistical Aggregation, Collaborative Refinement, Competitive Debate, and Final Adjudication, respectively.

overhead during scaling; [Qian *et al.*, 2025] reveals the bottlenecks of fully connected topologies.

Environment-Mediated. Environment-mediated structures replace point-to-point communication with a shared environment, enabling agents to collaborate indirectly by reading and writing shared global states, thereby reducing topological complexity. [Piao *et al.*, 2025] drives group behavior through social environments, while [Zhang *et al.*, 2025a] relies on system states and logs to coordinate actions. This paradigm has strong decoupling but relies on environment consistency. [Zhang *et al.*, 2025b] mitigates this through structured shared memory.

Dynamic Scheduling

Dynamic scheduling adjusts collaboration paths or structures through runtime decisions. By allocating computational and communication resources on demand during execution, this paradigm mitigates the redundant overhead of static processes, thereby improving system efficiency and adaptability.

Conditional Routing. Conditional routing dynamically selects agents or tools to participate in collaboration at runtime based on task semantics or intermediate states, avoiding re-

dundant computation from static flows. Its core mechanism typically relies on vector similarity [Wu *et al.*, 2025], rule triggering [Rong *et al.*, 2025], or confidence assessment for on-demand activation. This mechanism balances efficiency and performance but is highly sensitive to routing accuracy; incorrect decisions may amplify bias.

Dynamic Graph. Dynamic graph scheduling adjusts connections between agents during collaboration to adapt to task stages and contribution differences, reducing redundant interactions. Its core lies in restructuring the communication structure based on real-time performance, thereby enhancing overall system efficiency and scalability. [Liu *et al.*, 2024] characterizes the dynamic evolution of collaborative relationships, while [Zhang *et al.*, 2025d] and [Wang *et al.*, 2025d] reduce overhead through structure pruning and communication sparsification, though this potentially may weaken global information integration capabilities.

Automated Construction

Automated construction transforms LLM-MAS structure and collaboration flow design into a system self-generation problem, reducing reliance on manual orchestration and introduc-

ing new trade-offs between flexibility and stability.

Search-based. Such approaches regard the construction of multi-agent systems as a structural search problem, exploring efficient collaborative structures within predefined topological spaces. Their search strategies include graph search for discrete structures [Zhang *et al.*, 2025e] and optimization of continuous probability distributions [Zhang *et al.*, 2025c]. These methods can reduce human design biases, but are still limited by the combinatorial explosion of the search space and high evaluation costs.

Code Generation-based. This type of methods regards MAS construction as a code generation task, utilizing LLM to map natural language requirements into executable programs. Relevant studies have explored end-to-end MAS code generation [Ye *et al.*, 2025b], structured modeling based on Finite State Machines (FSM) [Zhang *et al.*, 2025h], and formal verification based on NuSMV [Agrawal *et al.*, 2025]. These methods have the advantages of interpretability and reproducibility, but are limited by the expressive boundaries of code and the reasoning ability of models.

Self-Organizing Dynamic Orchestration. Approaches in this category target scenarios with open environments and continuously changing tasks, transforming MAS design from an offline construction problem to a runtime governance issue. These systems can not only initialize role configurations and collaborative architectures based on task requirements [Li *et al.*, 2025b], but also dynamically generate new roles, adjust team boundaries, and evolve interaction topologies based on execution feedback [Wang *et al.*, 2025b]. When necessary, they can even introduce new members or tools from the external environment [Chen *et al.*, 2025a]. In addition, some studies have explored learnable online orchestration strategies [Dang *et al.*, 2025] to optimize dynamic paths. While such methods improve adaptability, they also pose challenges in terms of controllability and stability.

2.2 Interaction

This section focuses on the modes of information exchange and the mechanisms of collective decision-making.

Communication Paradigms

This section systematically categorizes communication paradigms in LLM-MAS into three types: natural language, structured language, and implicit vector.

Natural Language Communication. This paradigm utilizes unstructured, natural language as the primary carrier for information exchange among agents. Leveraging the general semantic capabilities of language models, it facilitates low-barrier collaboration but is constrained by context window limitations and expression ambiguity in large-scale scenarios.

1) Full-context dialogue flow. This mode maintains complete interaction histories to preserve role consistency and contextual coherence, typically employed in anthropomorphic interaction tasks such as role-playing and social simulation [Zhang *et al.*, 2025g]. **2) Reasoning-embedded communication.** This mode embeds the sender’s reasoning logic,

decision rationale, and step-by-step justifications into inter-agent communication [Zhang *et al.*, 2025f], enabling the recipient to verify the logical derivation path. **3) Compressed and summarized communication.** This mode reduces context and token overhead in long-horizon collaboration by compressing communication content [Wang *et al.*, 2025a] or condensing it into shared memory [Zhang *et al.*, 2025b].

Structured Language Communication. This paradigm does not exclude natural language text but anchors semantics to structured rules via predefined schemas. This enables agents to achieve controllability and interoperability without relying solely on natural language conventions. **1) Collaboration protocol specifications.** Targeted at inter-agent interaction, these specifications aim to achieve semantic alignment through standardized communication headers and payload formats. [Liao *et al.*, 2025] introduces the A2A protocol, explicitly constraining inter-agent communication via standardized JSON formats to structure task assignment, state feedback, and result delivery. **2) Operation interface specifications.** Targeted at agent interactions with tools or environments, these specifications map high-level reasoning results into executable atomic operations through structured instructions and parameterized action spaces. Related research mitigates natural language ambiguity and reduces execution instability via unified tool invocation and action interfaces [Yamada *et al.*, 2025; Agashe *et al.*, 2025; Zhang *et al.*, 2025a].

Implicit Vector Communication. This paradigm bypasses discrete natural language channels to exchange semantic information within the model’s latent representation space, aiming to mitigate information loss and computational overhead associated with symbolic encoding-decoding. Theoretical research has demonstrated the feasibility of sharing thought states between agents, revealing that precise semantic alignment across model representation spaces is achievable without discrete symbolic mediation [Zheng *et al.*, 2025b]. In engineering implementations, some methods lower interaction to the internal states of attention mechanisms, achieving high-bandwidth, low-latency implicit semantic transfer via cross-model reuse and alignment of the KV-Cache. This mechanism effectively supports real-time online multi-agent collaboration [Fu *et al.*, 2025; Ye *et al.*, 2025a].

Collective Decision-Making

This section systematically reviews four representative mechanisms in LLM-MAS for integrating dispersed judgments into consistent decisions.

Statistical Aggregation. Statistical aggregation smooths variance and hallucination risks in individual reasoning by integrating outputs from multiple independent agents. Beyond basic Majority Voting, research has introduced confidence-based weighting [Tang *et al.*, 2025] and consistency-constrained node filtering [Zheng *et al.*, 2025a] to further enhance decision stability and noise resistance.

Collaborative Refinement. Collaborative refinement focuses on constructive iterative optimization, refining preliminary results into high-quality outcomes through positive feedback loops and complementary division of labor among

agents. Common forms include reviewer agents providing multi-round improvement suggestions to generators [Yamada et al., 2025], slow-thinking agents retrospectively calibrating the reasoning paths of fast-thinking agents [Zhang et al., 2025f], and the establishment of additional supervision teams [Rong et al., 2025] or evolutionary teams [Hu et al., 2025b].

Competitive Debate. Competitive debate facilitates the gradual formation of correct judgments through the collision of conflicting viewpoints by introducing agents with differing stances or roles, often utilizing “tit-for-tat” adversarial mechanisms [Liang et al., 2024]. This approach not only significantly expands reasoning boundaries in complex tasks [Li et al., 2025a] but also resolves latent cognitive stances in decision-making games [Takano et al., 2025]. Furthermore, multi-dimensional debate architectures [Feng et al., 2025] have been proven to eliminate individual biases through multi-perspective cross-examination, achieving higher-quality consensus in complex tasks.

Final Adjudication. Final adjudication makes conclusive determinations on collective candidate results through non-negotiable criteria, such as rule verifiers or judge agents. This mechanism is common in formal specification-based verification [Agrawal et al., 2025], judge-based review [Bi et al., 2026], and authoritative guideline constraints in high-stakes domains [Xiao et al., 2025].

2.3 Evolution

This section focuses on how LLM-MAS continuously evolve collaborative capabilities through experience via different levels of evolutionary mechanisms.

Prompt-level Evolution

Unlike static prompt engineering, prompt-level evolution treats natural language prompts as editable hyperparameters, constructing an automated closed loop from execution feedback to updates. Addressing diverse optimization objectives, evolutionary mechanisms exhibit various forms: related research maps execution errors to text-level gradient signals, performing directed updates on agent collaboration prompts and behavioral instructions via textual backpropagation mechanisms [Hu et al., 2025b; Ma et al., 2025]. [Tian et al., 2025] conducts multi-round evaluation and restructuring of role definitions during the system construction phase. Furthermore, some studies leverage Monte Carlo search [Zhang et al., 2025e] or self-repair mechanisms [Wang et al., 2025b] to achieve adaptive evolution at the system architecture level.

Memory-level Evolution

Memory-level evolution aims to simulate the cognitive process of human skill acquisition from experience. By dynamically accumulating, updating, and optimizing memory, systems continuously enhance multi-agent collaboration or task-solving capabilities. [Fan et al., 2025] consolidates execution feedback into reusable long-term memory, distinguishing between intra-task experience and cross-task constraints to reduce redundant trial-and-error. At the operational execution level, certain methods focus on accumulating operational experience and behavioral patterns, supporting agents in adaptively optimizing actions based on his-

torical records [Wang et al., 2025c]. Furthermore, structured shared memory is utilized for collective cognitive accumulation, abstracting interaction records into high-level knowledge via hierarchical or graph structures to facilitate experience generalization and transfer [Zhang et al., 2025b; Rong et al., 2025].

Parameter-level Evolution

Parameter-level evolution aims to internalize collaborative capabilities into native model intelligence via gradient updates, thereby overcoming the performance bottlenecks of prompt engineering in practice. Existing research primarily focuses on reinforcement learning-based co-evolution. Addressing challenges such as credit assignment and gradient blocking in collaboration, [Zhao et al., 2025] and [Hong et al., 2025] introduce Co-PPO and hierarchical GRPO algorithms, respectively, achieving collaborative policy updates within flat and leader-follower architectures. [Park et al., 2025] and [Chen et al., 2025b] further construct unsupervised evolutionary closed loops based on validator feedback and self-play, driving the autonomous emergence of collaborative strategies among agents.

3 Frontier Applications

Motivated by diverse application scenarios, this section systematically reviews the major frontiers of LLM-MAS across three complementary dimensions, as summarized in Table 2.

3.1 AI for Humans

This section focuses on leveraging LLM-MAS to simulate expert teams, facilitating the decomposition and execution of complex tasks faced by humans.

Software Engineering. Addressing challenges associated with intricate logic and long-range dependencies in software engineering, LLM-MAS decomposes tasks through collaborative role simulation, such as architects, programmers, and testers. In code generation, [Seo et al., 2025] achieves automated conversion from research papers to code repositories. For defect rectification, [Li et al., 2025a] introduces a debate mechanism to conduct multi-perspective code reviews. In software testing, [Hu et al., 2025b] constructs a self-evolving collaboration network to continuously enhance software quality through feedback.

Scientific Research. Scientific research faces challenges in interdisciplinary integration, rigorous verification, and high trial-and-error costs. LLM-MAS typically constructs digital research teams composed of principal investigators, executors, and reviewers to automate the pipeline from hypothesis generation to result production. Existing works include full-process autonomous research agents [Yamada et al., 2025], iterative hypothesis verification [Weng et al., 2025a], and interdisciplinary collaborative verification in biomedicine [Swanson et al., 2025] and chemistry [Yang et al., 2025a].

Healthcare. Given the requirements for accuracy and evidence traceability in medical decision-making, LLM-MAS typically simulates the Multi-Disciplinary Team (MDT) paradigm. Recent studies involve multi-round debate mechanisms to resolve diagnostic disagreements [Xiao et al., 2025],

Motivation	Domain	Work
AI for Humans	Software Engineering	[Seo <i>et al.</i> , 2025; Li <i>et al.</i> , 2025a; Hu <i>et al.</i> , 2025b]
	Scientific Research	[Yamada <i>et al.</i> , 2025; Weng <i>et al.</i> , 2025a; Swanson <i>et al.</i> , 2025; Yang <i>et al.</i> , 2025a]
	Healthcare	[Xiao <i>et al.</i> , 2025; Liu <i>et al.</i> , 2025; Sun <i>et al.</i> , 2025; Zhu <i>et al.</i> , 2025]
	GUI Interaction	[Agashe <i>et al.</i> , 2025; Zhang <i>et al.</i> , 2025a; Zhou <i>et al.</i> , 2025; Wang <i>et al.</i> , 2025c]
	Information Retrieval	[Li <i>et al.</i> , 2025c; Liao <i>et al.</i> , 2025; Wu <i>et al.</i> , 2025; Chang <i>et al.</i> , 2025]
	Content Creation	[Pang <i>et al.</i> , 2025; Ji <i>et al.</i> , 2025; Rong <i>et al.</i> , 2025; Xu <i>et al.</i> , 2025]
AI for the World	Societal Simulation	[Piao <i>et al.</i> , 2025; AL <i>et al.</i> , 2024; Zhang <i>et al.</i> , 2025g]
	Embodied AI	[Zu <i>et al.</i> , 2025; Chai <i>et al.</i> , 2025; Yang <i>et al.</i> , 2025b]
	Economics	[Hao and Xie, 2025; Zhao <i>et al.</i> , 2024; Chu <i>et al.</i> , 2025; Xia <i>et al.</i> , 2025]
	Psychology	[Zhang <i>et al.</i> , 2025g; Weng <i>et al.</i> , 2025b; Luo and Laban, 2025]
AI for AI	Data Engineering	[Yue <i>et al.</i> , 2025; Yang <i>et al.</i> , 2025c; Hegazy <i>et al.</i> , 2025; Mu <i>et al.</i> , 2026; Lu <i>et al.</i> , 2025]
	System Optimization	[Tian <i>et al.</i> , 2025; Wang <i>et al.</i> , 2025a; Zhao <i>et al.</i> , 2025; Zhang <i>et al.</i> , 2025d]
	Automated Evaluation	[Vu <i>et al.</i> , 2025; Bi <i>et al.</i> , 2026; Feng <i>et al.</i> , 2025; Hu <i>et al.</i> , 2025a]
	Security Defense	[Kavathekar <i>et al.</i> , 2025; Miao <i>et al.</i> , 2025; Zheng <i>et al.</i> , 2025a]

Table 2: Applications of LLM-MAS categorized by motivation and domain.

clinical imaging reasoning [Liu *et al.*, 2025], the synthesis of large-scale medical reasoning samples [Sun *et al.*, 2025], and the establishment of systematic evaluation benchmarks for multi-agents in diverse medical tasks [Zhu *et al.*, 2025].

GUI Interaction. To address single-model limitations in planning and perception, LLM-MAS typically adopts a collaborative paradigm that decouples high-level strategic planning from low-level operational execution. Representative works include desktop manipulation combining visual specialists [Agashe *et al.*, 2025], cross-application synergy via deep system integration [Zhang *et al.*, 2025a], mobile-based long-horizon task execution utilizing retrieval augmentation [Zhou *et al.*, 2025], and interactive adaptability enhancement based on self-evolving memory [Wang *et al.*, 2025c].

Information Retrieval. Addressing the challenges of massive heterogeneous data and long-tail queries, LLM-MAS reconstructs the retrieval process as intent-driven deep exploration. Relevant research includes dynamic orchestration adapted to query complexity [Li *et al.*, 2025c], modal data source interoperability [Liao *et al.*, 2025], expert matching based on vector routing [Wu *et al.*, 2025], and content filtering leveraging collaborative filtering to enhance the signal-to-noise ratio [Chang *et al.*, 2025].

Content Creation. Targeting the difficulties of semantic alignment and temporal synchronization in multi-modal generation, systems introduce layered generation and supervision mechanisms, ensuring global consistency through specialized roles such as parsing, illustration, and dubbing. Recent studies include editable poster generation [Pang *et al.*, 2025], educational content creation [Ji *et al.*, 2025], and immersive story video generation [Xu *et al.*, 2025]. Additionally, [Rong *et al.*, 2025] employs a dual-layer architecture to guarantee spa-

tiotemporal consistency in multi-audio generation.

3.2 AI for the World

This section focuses on leveraging LLM-MAS to simulate real-world dynamics. By constructing virtual social, economic, and physical environments, LLM-MAS facilitates the deduction and prediction of complex system evolution.

Societal Simulation. Social simulation aims to model micro-level group interactions and macro-level societal evolution. LLM-MAS employs generative sandboxes to endow agents with human-like cognitive architectures, thereby achieving high-fidelity simulation. Representative studies range from the micro to the macro level, including the simulation of interpersonal interactions [Zhang *et al.*, 2025g], city-scale behavioral simulation [Piao *et al.*, 2025], and the spontaneous evolution of law and civilization [AL *et al.*, 2024].

Embodied AI. To achieve perception, planning, and collaborative action within physical or virtual environments, LLM-MAS establishes a closed-loop paradigm bridging environmental perception and collaborative decision-making. Recent works encompass non-simulation reward optimization that transforms collaboration into tree search [Zu *et al.*, 2025], causal planning in open-world settings [Chai *et al.*, 2025], and distributed collaboration based on adaptive experience sharing [Yang *et al.*, 2025b].

Economics. By simulating heterogeneous agents as digital twins, LLM-MAS deduces macro-policies from micro-behaviors to capture complex economic dynamics. Research directions include deducing the dynamic impacts of policies on economic systems [Hao and Xie, 2025], analyzing price wars and collusion in competitive markets [Zhao *et al.*, 2024], simulating marketing strategies [Chu *et al.*, 2025], and exploring collaborative filtering in recommender systems [Xia *et al.*, 2025].

Psychology. Psychology research focuses on dissecting individual cognitive processes and micro-psychological mechanisms. By simulating metacognition and social interaction, LLM-MAS reveals the isomorphism between machine behavior and human psychology. In the realm of social cognition, [Zhang *et al.*, 2025g] introduces a metacognitive architecture to simulate human Theory of Mind and moral norms. At the group psychology level, [Weng *et al.*, 2025b] uncovers conformity tendencies within group interactions. Furthermore, [Luo and Laban, 2025] establishes a multi-agent evaluation framework designed to detect psychological risks.

3.3 AI for AI

This section focuses on leveraging multi-agent collaboration to bolster the foundational capabilities of LLM-MAS itself. LLM-MAS can be utilized to feed back into itself in processes such as data generation, automated evaluation and security governance.

Data Engineering. Addressing the scarcity of high-quality data and the prohibitive costs of annotation, LLM-MAS simulates human workflows for data production and processing. In data generation, [Yue *et al.*, 2025] synthesizes instruction

data by simulating teacher-student interactions, while collaborative multilingual agents have proven effective in constructing high-quality code instruction sets [Yang *et al.*, 2025c]. In the realm of automated annotation, [Hegazy *et al.*, 2025] achieves high-precision labeling for enterprise-scale data, and [Mu *et al.*, 2026] overcomes bottlenecks in low-resource entity recognition. Furthermore, regarding data structuring, [Lu *et al.*, 2025] automates the enrichment of knowledge graphs.

System Optimization. Distinct from solving external application tasks, system optimization aims to transform inter-agent interaction feedback into optimization signals, iteratively upgrading architectures, parameters, and communication patterns. Recent works apply this to multi-agent system construction [Tian *et al.*, 2025], collaborative reasoning architecture enhancement [Wang *et al.*, 2025a], parameter-efficient multi-model training [Zhao *et al.*, 2025], and communication redundancy pruning for efficiency optimization [Zhang *et al.*, 2025d].

Automated Evaluation. LLM-MAS performs automated evaluation via digital juries simulating human debate and scrutiny. In general domains, [Vu *et al.*, 2025] mimics multi-dimensional human judgment regarding generated content, while JudgeBoard [Bi *et al.*, 2026] validates that collaboration among small models can achieve high-precision reasoning assessment. For domain-specific evaluation, [Feng *et al.*, 2025] realizes fine-grained assessment of machine translation. Additionally, related research introduces adaptive stability detection to optimize the trade-off between evaluation cost and reliability [Hu *et al.*, 2025a].

Security Defense. Safety and defense initiatives are dedicated to enhancing system robustness and resistance to attacks through collective collaboration mechanisms. [Kavathekar *et al.*, 2025] establishes a comprehensive benchmark covering diverse attack categories, conducting adversarial risk testing based on multi-agent interaction scenarios. Regarding active defense, [Miao *et al.*, 2025] propose unsupervised malicious behavior detection mechanisms. Furthermore, in system resilience research, related work quantifies critical metrics for consensus safety [Zheng *et al.*, 2025a].

4 Challenges and Opportunities

While LLM-MAS demonstrate significant potential in complex tasks, transitioning from experimental settings to practical deployment faces fundamental challenges. We identify obstacles and research opportunities impeding system scalability and real-world application.

4.1 Hallucination Propagation Control

Individual hallucinations within interaction chains are prone to amplification into collective biases, and static validation mechanisms fail to arrest error accumulation during dynamic collaboration. The core future direction involves developing cognitive provenance technologies and full-lifecycle trustworthy verification frameworks. By introducing external knowledge anchors and dynamic error correction mechanisms, we can intercept the cascading diffusion of errors during long-chain reasoning in real time.

4.2 Communication Efficiency

Fully connected natural language communication faces scaling bottlenecks [Qian *et al.*, 2025]. The key to a breakthrough lies in the transition toward implicit vector representations, leveraging latent space for semantic exchange combined with sparse interaction to drastically reduce token consumption and decoding latency. Constructing such high-bandwidth architectures is the core pathway to achieving real-time collaboration in large-scale agent clusters.

4.3 Evolutionary Capability

As indicated in Table 1, most current LLM-MAS lack self-evolutionary capabilities. A central opportunity lies in constructing a three-level evolutionary mechanism encompassing “Prompt-Memory-Parameter”. Simultaneously, frontier research has begun exploring adaptive architectural evolution, enabling LLM-MAS to dynamically restructure collaborative topologies and role divisions based on task demands.

4.4 Emergence of Collective Intelligence

Current LLM-MAS lack the capability to predict or attribute emergent behaviors, making it difficult to distinguish between beneficial emergence and harmful deviations [Weng *et al.*, 2025b]. Research emphasis must shift toward establishing mathematical representations of collective behavior and principle-based guidance mechanisms. By mapping emergent capabilities to human values, we can transform unpredictability into controllable productivity.

4.5 Evaluation Framework

Current evaluations are largely confined to individual capabilities or final outputs, lacking unified standards for the collaborative process. There is an urgent need to construct multi-dimensional benchmarks covering communication overhead, strategic consistency, and robustness. This facilitates a shift in the evaluation paradigm from a purely result-oriented approach to one that weighs both process and outcome, providing an objective basis for comparing heterogeneous systems.

4.6 Security Governance

Multi-agent interactions introduce novel risks, including cascading propagation of malicious instructions and emergent collusion. Effective governance requires distributed defense mechanisms and clear accountability frameworks. Future work should emphasize collective value alignment and end-to-end auditing to ensure the trustworthiness and controllability of decentralized decision-making systems.

5 Conclusion

This paper constructs a three-dimensional framework of “Structure-Interaction-Evolution”, systematically reviewing the core paradigms and recent advancements of LLM-MAS while categorizing their broad application frontiers. Despite facing challenges such as hallucination propagation and safety governance, LLM-MAS remain an essential pathway to breaking single-agent bottlenecks and advancing toward Artificial General Intelligence. This paper aims to provide a solid conceptual foundation for future architectural innovation and theoretical research.

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